Prabhavit Gupta

AI Engineer: Multi-Step Agents, MCP Architecture, RAG & Production Pipelines

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Summary

AI Software Engineer with 3 years of commercial production experience architecting autonomous multi-step LLM agents, Model Context Protocol (MCP) integrations, and automated document/data extraction pipelines. Proven track record deploying production AI systems that replace manual administrative and operational friction, handling unstructured records (contracts, payroll-adjacent documents, invoices) for 4,500+ active enterprise users. Expert in building resilient agent orchestration, hybrid vector/keyword RAG architectures, and deterministic validation schemas to guarantee reliability in mission-critical business environments. Based in Venice, Italy with permanent EU work authorization (Partita IVA / Italian employment contract ready).

Experience

FlowSpace.co • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Engineered production multi-step AI agents and backend data pipelines automating unstructured document extraction (invoices, forms, identity records) for 4,500+ active enterprise clients.
- Implemented Model Context Protocol (MCP) tool interfaces, enabling agents to dynamically discover schemas, query structured databases, and execute validated actions with robust error recovery.
- Architected enterprise RAG and semantic retrieval systems combining PostgreSQL pgvector embeddings, dense retrieval, and metadata filtering for high-precision knowledge grounding.
- Built deterministic guardrails and Pydantic schema validation layers around model outputs, preventing hallucinated or malformed records from reaching downstream operational systems.
- Implemented asynchronous task workers and Celery/queue processing with idempotency keys and exponential retries to guarantee zero-loss execution under heavy traffic spikes.
- Automated continuous integration and deployment pipelines using Docker containerization, pytest suites, and GitHub Actions across AWS cloud infrastructure.
- *Tools: Python, LangChain, PostgreSQL, MCP, Docker, AWS, Hugging Face, REST APIs, Linux*

Urlink • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Developed automated Python data ingestion routines and entity transformation pipelines connecting heterogeneous third-party APIs to central relational stores.
- Designed automated schema validation checks and data normalization layers to enforce data integrity across multi-tenant enterprise accounts.
- Collaborated directly with product and operations stakeholders to translate business requirements into technical interface specifications and monitoring alerts.
- *Tools: Python, REST APIs, PostgreSQL, Supabase, TypeScript, Linux*

Independent Technical Consultant • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Developed custom Python automation workers, webhook listeners, and document parsing agents for automated information extraction and database ingestion.
- Established automated regression test harnesses and synthetic evaluation benchmarks to measure agent tool-calling accuracy across prompt iterations.
- Authored detailed system documentation, error-handling playbooks, and schema contracts for client engineering teams.
- *Tools: Python, LangChain, Webhooks, REST APIs, Supabase, Linux*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Integrated LLM endpoints and automated notification workflows, establishing data validation and consistency checks across core user onboarding flows.
- *Tools: Python, OpenAI API, Sendgrid, REST APIs*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** Permesso di Soggiorno UE per Soggiornanti di Lungo Periodo (Italy), unrestricted work authorization in Italy and EU (CCNL or Partita IVA). No visa sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (working beginner).
- **Availability:** Available immediately for full-time remote roles in Italy.

Core Skills

- **AI Agents & LLM Systems:** Multi-step agent orchestration, Model Context Protocol (MCP), LangChain, tool calling, prompt engineering, agentic error recovery.
- **RAG & Knowledge Bases:** Vector search, embeddings, document chunking, semantic retrieval, RAG evaluation, knowledge base maintenance and versioning.
- **Backend Engineering:** Python 3, asynchronous programming, clean modular architecture, REST APIs, Pydantic, schema validation, pytest.
- **Data Systems & Storage:** PostgreSQL, relational schema design, SQL query optimization, data cleansing and ingestion pipelines, Supabase, Redis caching.
- **Cloud & DevOps:** Docker containerization, AWS, Linux system administration, Git, GitHub Actions CI/CD, task queues, logging and telemetry.